TUTORIAL - 2

Data Types in Biomedical Informatics

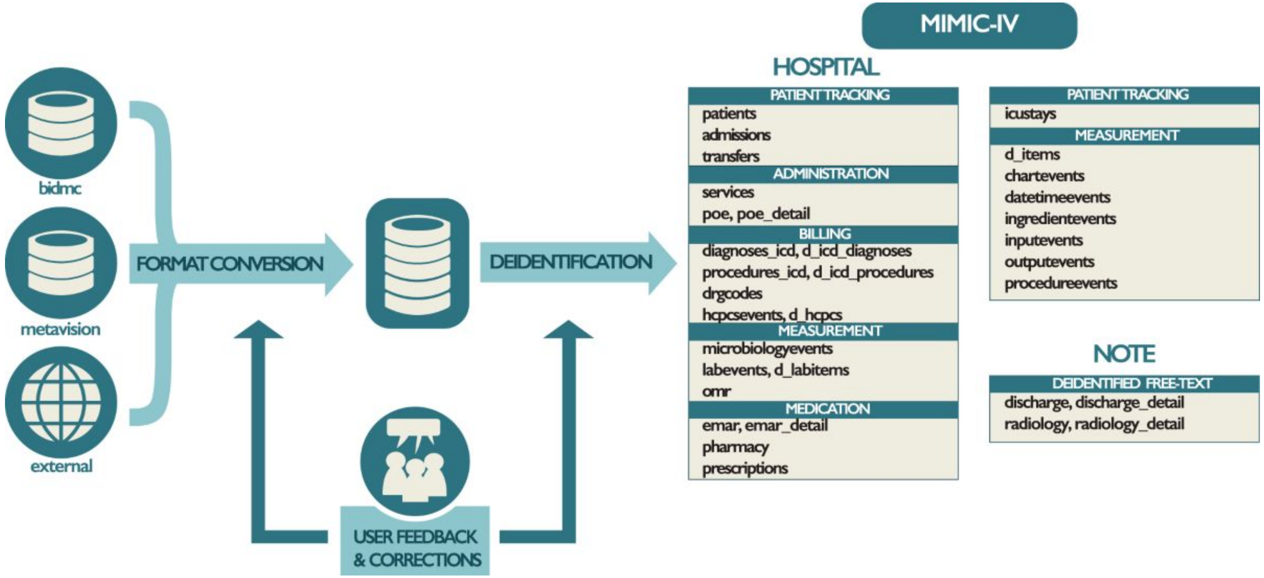
Understanding Data Before Building AI Models

Learning Objectives

- Understand major biomedical data types
- Know how data type affects ML model choice
- Understand challenges in biomedical data
- Be able to choose data type for your project

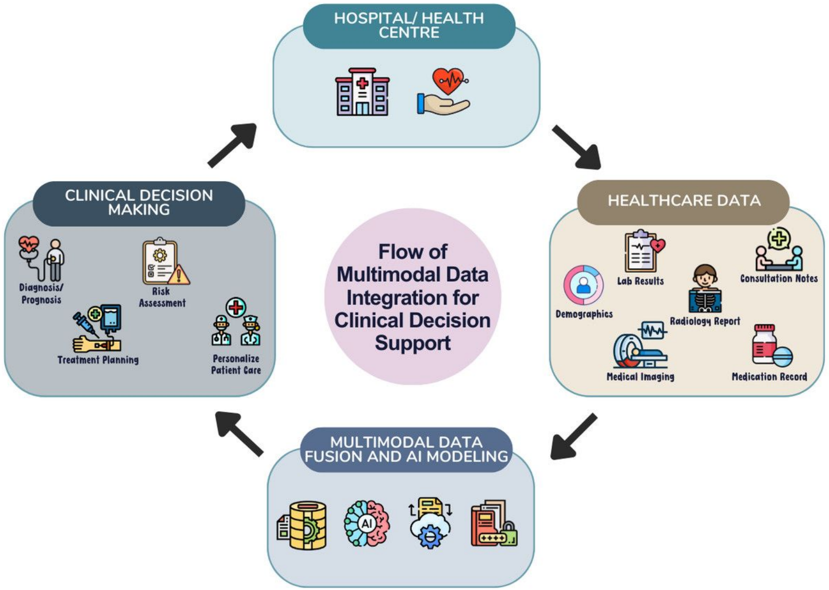
Recap of MIMIC

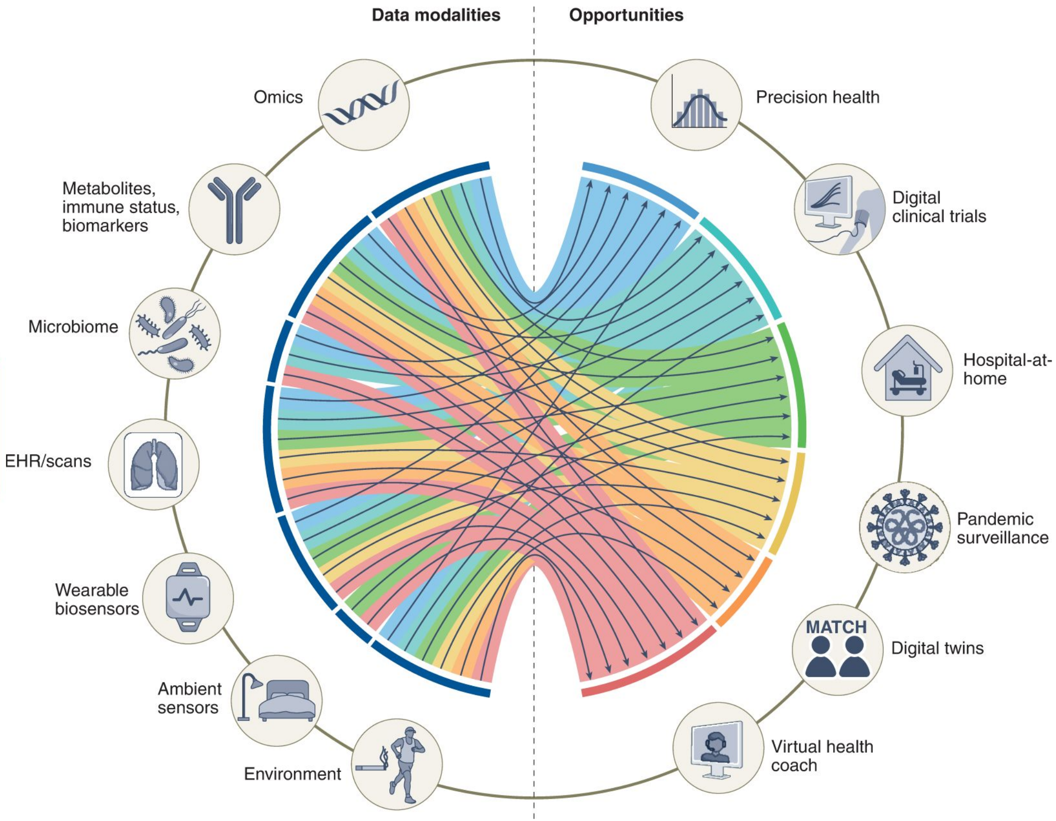
- Medical Information Mart for Intensive Care (MIMIC)
- Large, open-access ICU database
- Developed by MIT & Beth Israel Deaconess Medical Center
- Used worldwide for clinical & ML research



The Biomedical Data Ecosystem

Biomedical data is **multimodal**



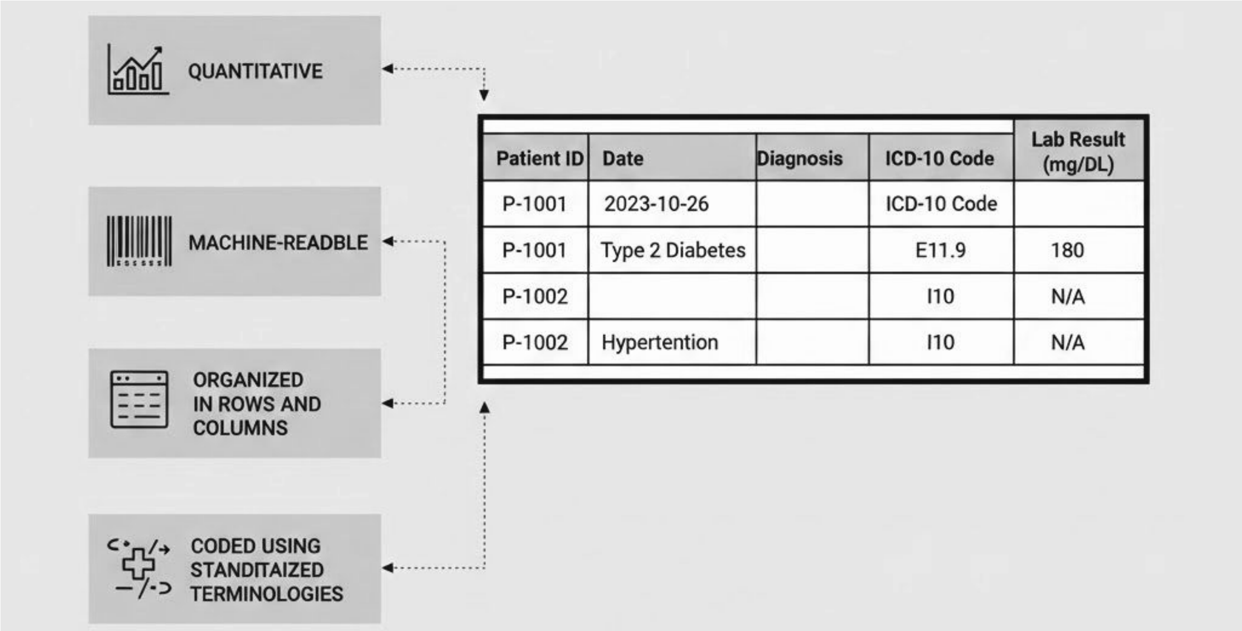


Source: Acosta, J.N., Falcone, G.J., Rajpurkar, P. *et al.* Multimodal biomedical AI. *Nat Med* 28, 1773–1784 (2022). https://doi.org/10.1038/s41591-022-01981-2

Structured Clinical Data in Biomedical Informatics

- ➢ Highly organized, machine-readable information—such as demographics, vitals, lab results, and diagnostic codes (ICD-10)—stored in databases like Electronic Health Records (EHRs)
- ➢ Often high dimensional but interpretable.
- ➢ Important Concept:

  "***Models trained on one hospital may fail in another."***



Patient ID	Date	Diagnosis	ICD-10 Code	Lab Result (mg/DL)
P-1001	2023-10-26		ICD-10 Code	
P-1001	Type 2 Diabetes		E11.9	180
P-1002			I10	N/A
P-1002	Hypertention		I10	N/A

Common Sources

- Electronic Health Records (EHR)
- ICU databases
- Hospital Information Systems
- Clinical registries

Well-known examples:

- MIMIC
- UK Biobank
- eICU Collaborative Research Database

Examples of Structured Variables

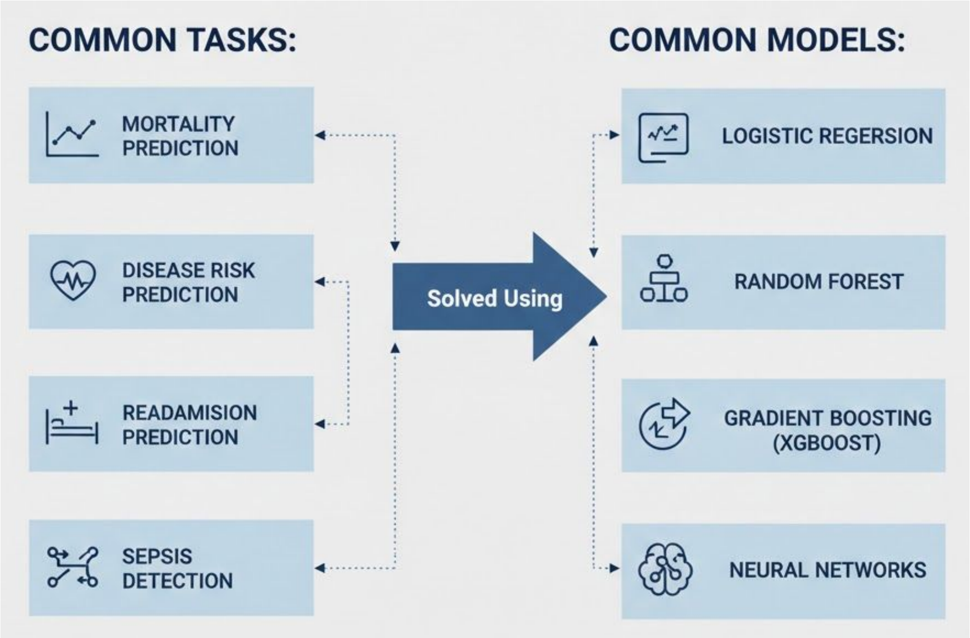
- Demographics (Age, Sex, Ethnicity)
- Vital signs (Heart rate, Blood pressure)
- Laboratory values (Glucose, Hemoglobin, Creatinine)
- Diagnosis codes (ICD-10)
- Medications
- Length of stay
- Mortality outcome

Key Characteristics

- Structured & tabular
- Often contains missing values
- Class imbalance common
- Requires preprocessing
- Easier interpretability compared to imaging

Major Challenges

- Missing data (common in EHR)
- Coding inconsistencies
- Data bias
- Temporal leakage
- Generalization across hospitals

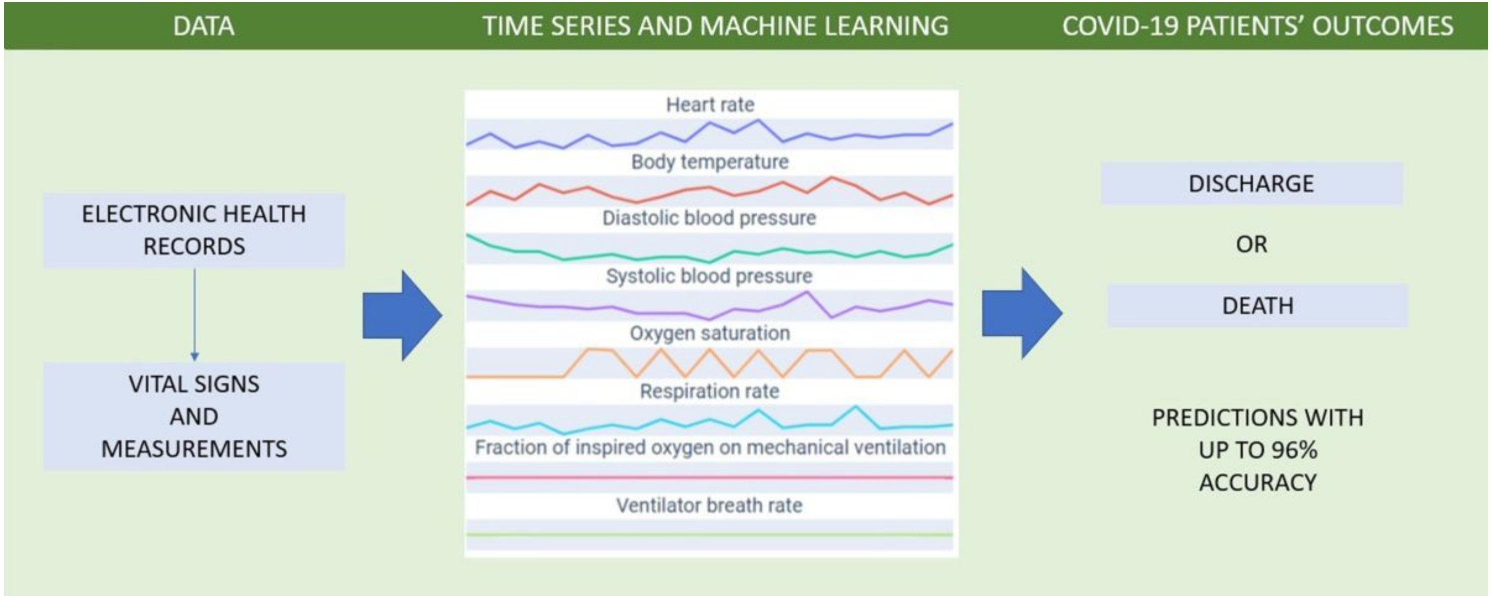


Time-Series Data in Biomedical Informatics

Time-series data refers to data points collected sequentially over time, where:

The order and timing of measurements matter.

Unlike tabular data (static snapshot), time-series captures **temporal dynamics**.



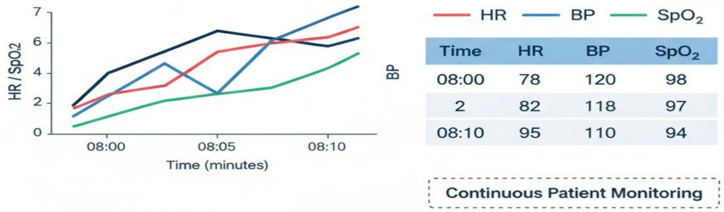
Time-series modeling helps in:

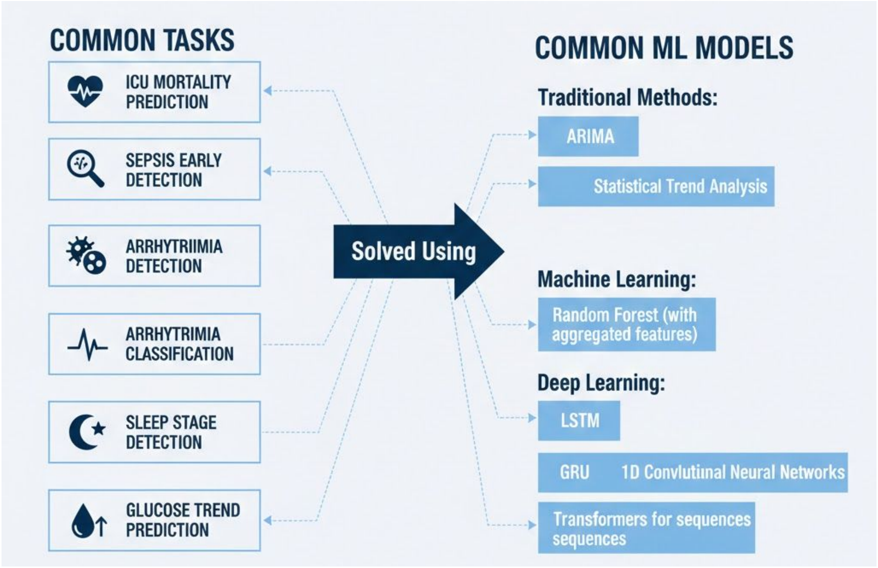
- Early warning systems
- Real-time ICU decision support
- Predicting deterioration before it becomes critical

Source: Rodrigues, D.S., Nastri, A.C.S., Magri, M.M. et al. Predicting the outcome for COVID-19 patients by applying time series classification to electronic health records. BMC Med Inform Decis Mak 22, 187 (2022). https://doi.org/10.1186/s12911-022-01931-5

Time-Series Data:

- Sequential
- Pattern changes over time
- Previous value influences next value





Major Challenges

- Irregular time intervals
- Missing data
- High noise
- Data leakage (using future info accidentally)
- Overfitting small datasets

Important Risk:

Temporal leakage → using future data during training → unrealistic performance.

Key Characteristics

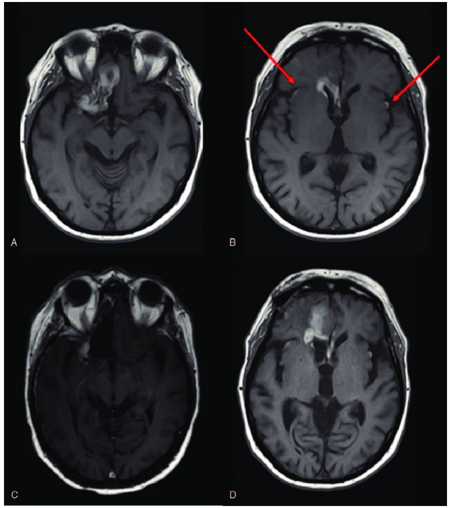
- Temporal dependency
- Variable sequence lengths
- Irregular sampling intervals
- Missing time points
- Noise in signals

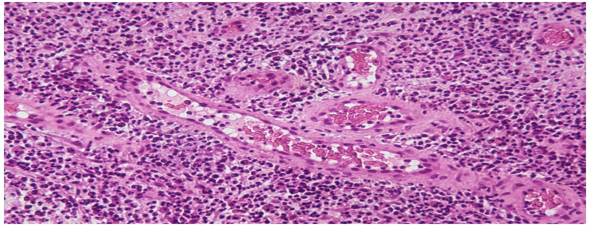
Medical Imaging Data in Biomedical Informatics

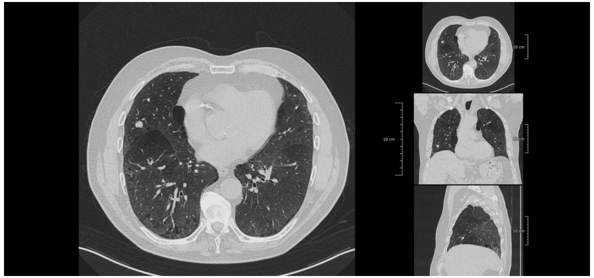
Medical imaging data refers to visual representations of internal body structures used for diagnosis, monitoring, and treatment planning.

Unlike tabular data:

- Images are high-dimensional
- Feature extraction is automatic (via deep learning)
- Interpretation often requires clinical expertise







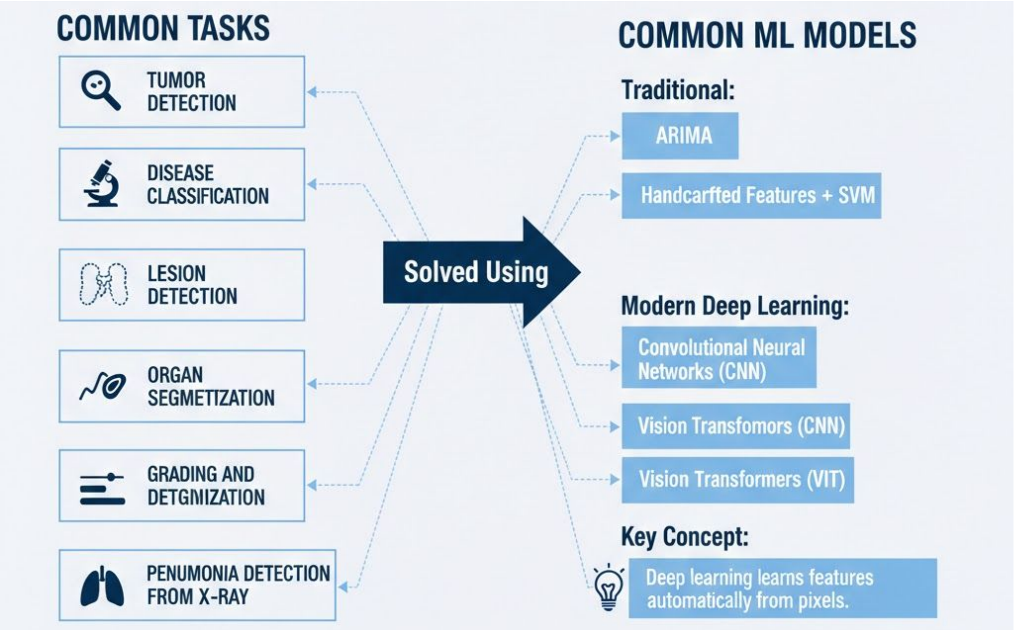
- MRI (Magnetic Resonance Imaging)
- CT (Computed Tomography)
- X-ray
- Ultrasound
- Histopathology (Whole Slide Images)

An image is:

- A matrix of pixel intensity values
- 2D or 3D (volumetric scans like MRI)
- Sometimes extremely large (WSIs can be several GBs)

Important Concept:

<u>Images contain spatial relationships — nearby pixels are related.</u>



Key Characteristics

- Very high dimensional
- Large storage requirements
- Requires preprocessing
- Often annotated by experts
- May require GPU for training

Major Challenges

- Large dataset requirement
- Data imbalance
- Annotation cost
- Overfitting on small datasets
- Generalization across hospitals/scanners
- Ethical and bias concerns

Genomic / Omics Data in Biomedical Informatics

Genomic / Omics data refers to large-scale biological measurements that capture molecular-level information about cells and tissues.

Unlike clinical data:

- Extremely high dimensional
- Biologically complex
- Often small sample size

Key Concept:

Features >> Samples

Type	Measures	Example
Genomics	DNA mutations	SNP analysis
Transcriptomics	RNA expression	RNA-seq
Proteomics	Protein abundance	Mass spectrometry
Epigenomics	DNA methylation	Methylation arrays

Data Structure

Typically stored in tabular format:

Rows → Patients
Columns → Thousands of genes (often 20,000+)

This creates a high-dimensional problem.

Important Concept: Curse of Dimensionality

When:

Number of features >> Number of samples

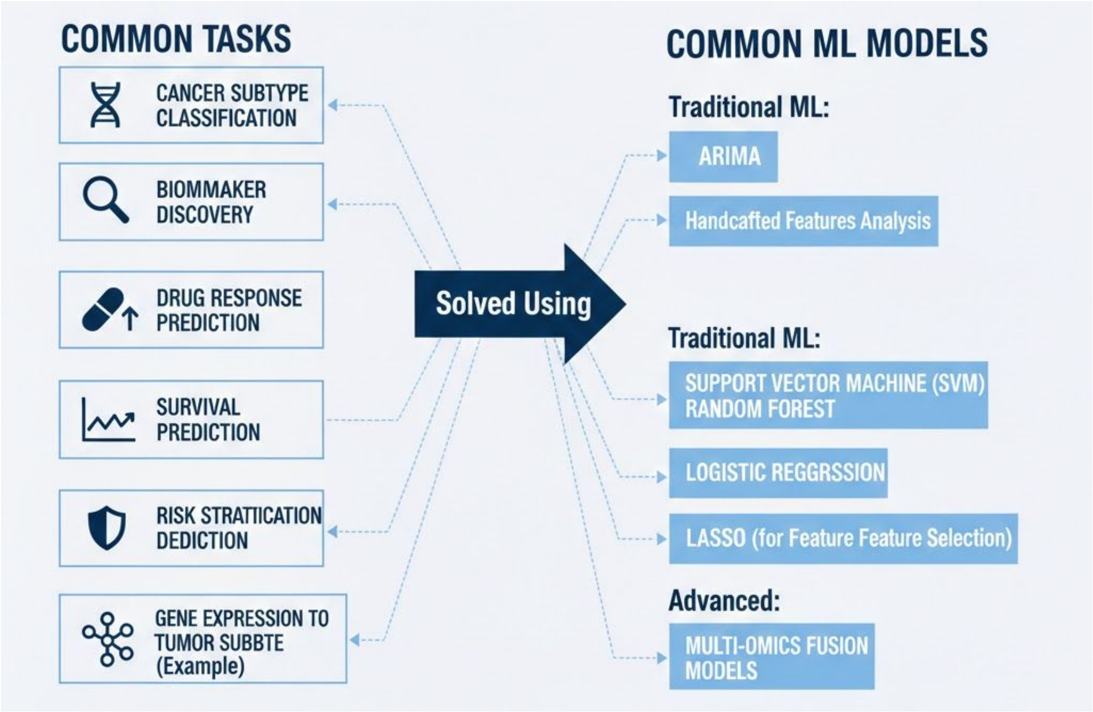
Problems:

- Overfitting
- Noise dominance
- Reduced generalization
- Hard biological interpretation

Feature selection becomes critical.

Major Challenges

- Small sample size
- Batch effects
- High noise
- Biological variability
- Interpretability requirement
- Reproducibility issues



WHY???

Reveals molecular mechanisms
Enables precision medicine
Helps identify biomarkers
Supports personalized treatment

Clinical Text Data & NLP in Biomedical Informatics

Clinical text data refers to unstructured, free-text medical documentation generated during patient care.

Common Sources of Clinical Text

- Discharge summaries
- Radiology reports
- Pathology reports
- Physician notes
- Nursing notes
- Operative notes

Well-known datasets:

- MIMIC (contains clinical notes)
- PhysioNet

WHY???

Structured data may say:

- Diagnosis code: Pneumonia

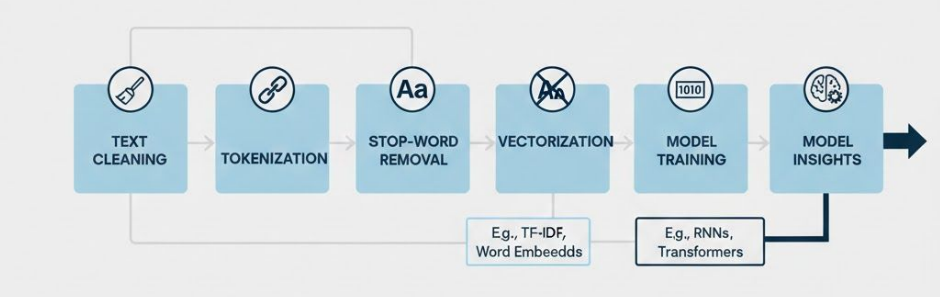
But clinical note may say:

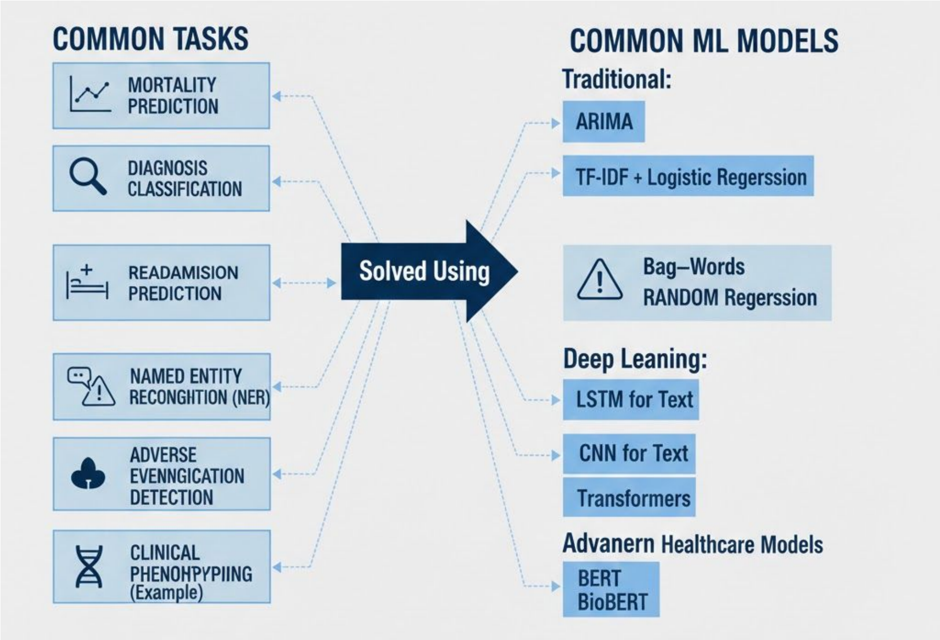
- "Patient presents with worsening bilateral infiltrates and shortness of breath."

Text contains:

- Clinical reasoning
- Context
- Severity
- Uncoded details

It often contains information not captured in structured fields.





Major Challenges in Clinical NLP

- Domain-specific language
- Limited labeled data
- Privacy regulations (HIPAA)
- Model hallucination risk
- Generalization across institutions

Important Concept: Context Matters

Clinical NLP can:

- Extract hidden information
- Automate report analysis
- Improve clinical decision support
- Enhance patient risk prediction

Comparison of Biomedical Data Types

Data Type	Example Source	Structure	ML Models	Data Size	Complexity
Structured Clinical	EHR, MIMIC	Tabular	Logistic Regression, RF, XGBoost	Moderate	Low–Moderate
Time-Series	ICU monitors, PhysioNet	Sequential	LSTM, GRU, 1D CNN	Moderate	Moderate
Imaging	MRI, CT, X-ray	Pixel matrix (2D/3D)	CNN, Vision Transformers	Large	High
Genomic / Omics	RNA-seq, Microarray	High-dimensional table	SVM, LASSO, RF	Small–Moderate	High
Clinical Text	Clinical notes	Unstructured text	BERT, Transformers	Large	High

Confusion Matrix

	Actually Positive (1)	Actually Negative (0)
Predicted Positive (1)	True Positives (TPs)	False Positives (FPs)
Predicted Negative (0)	False Negatives (FNs)	True Negatives (TNs)

TP → Correct disease detection

FP → False alarm

FN → Missed disease

TN → Correct rejection

Accuracy

- Measure a model's performance by quantifying the correctness of its predictions.
- Measures the proportion of total correctly predicted instances (both positive and negative) out of the total instances.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Limitations:
It can be deceptive if classes are imbalanced (e.g., 95% class A, 5% class B), as a model might achieve 95% accuracy by only predicting the majority class.

Precision

measures the accuracy of positive predictions by calculating the ratio of true positives to total predicted positives (True Positives / (True Positives + False Positives))

$$Precision = \frac{TP}{TP + FP}$$

Prioritize precision when the goal is to avoid false positives

"Of all items labeled as positive, how many were actually positive?"

Recall (Sensitivity)

The true positive rate (TPR), or the proportion of all actual positives that were classified correctly as positives, is also known as recall.

$$\text{Recall} = \frac{TP}{TP + FN}$$

The cost of false negatives is high (e.g., missing a disease).

Specificity

Specificity, or the True Negative Rate (TNR), measures a classification model's ability to correctly identify negative instances (non-events). It is defined as the proportion of actual negatives that are correctly identified

$$\text{Specificity} = \frac{\text{True negative}}{\text{True negative} + \text{false positive}}$$

In medical testing, it indicates how well a test excludes a disease for people who do not have it.

It prioritizes minimizing False Positives (Type I errors).

F1 Score

The F1 score is the harmonic mean of precision and recall, providing a balanced performance metric for classification models, especially with imbalanced datasets.

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

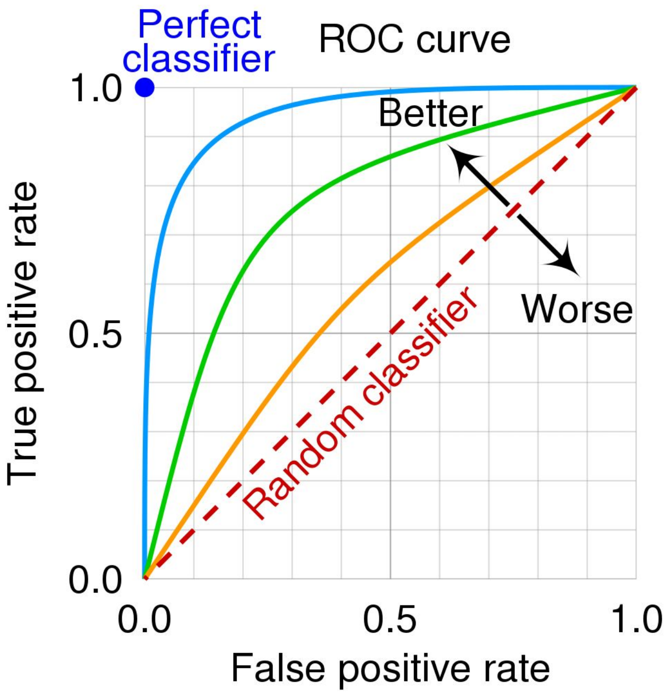
High F1: Indicates high precision and high recall.

AUC (Area Under ROC Curve)

- Key performance metric for binary classification models, representing the probability that a model ranks a random positive example higher than a random negative one.

- Ranging from 0 to 1, it measures a model's ability to distinguish between classes across all possible thresholds, where is random guessing and is perfect discrimination.

- The ROC curve itself plots the True Positive Rate (Sensitivity) on the y-axis against the False Positive Rate (specificity) on the x-axis.



From Data Type → Model → Evaluation Metric

Wrong evaluation → Wrong conclusion

Wrong conclusion → Risk to patients

Data Type	Typical Models	Key Evaluation Metrics	Why This Metric?
Structured Clinical	Logistic Regression, Random Forest, XGBoost	F1-score, AUC-ROC, Recall	Often imbalanced; need balanced performance
Time-Series	LSTM, GRU, 1D CNN	Recall, AUC, F1	Missing deterioration is dangerous
Medical Imaging	CNN, Vision Transformer	Sensitivity, Specificity, AUC	Need to detect disease without over-diagnosing
Genomic / Omics	Any LLMs (word2vec, BERT et .,), SVM, LASSO, RF	F1-score, AUC, Cross-validation stability	High dimensional, small sample size
Clinical Text (NLP)	BERT, Transformers	Precision, Recall, F1-score	Language ambiguity & class imbalance